Noise-to-Abundance Translation: Unsupervised Hyperspectral Unmixing Based on Diffusion Models

Hua Dong, Xiaohua Zhang, *Member, IEEE*, Hongyun Meng, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Hyperspectral unmixing is an important issue in the remote sensing field for identifying the constituent materials and their respective distributions in a scene. In recent years, a large number of data-driven HSI unmixing methods have been proposed. These methods reformulate the unmixing task as a transformation from spectral pixels to abundance. Diffusion models, a recently emerging class of generative models, have shown potential in converting noise into other meaningful representations. However, applying diffusion models to unsupervised hyperspectral unmixing faces the following challenges: stochasticity of the generative process, neglect of the quality and prior knowledge of the estimation of original data, and difficulty in solving the learning optimization problem. To address these issues, we propose a novel diffusion-based unsupervised hyperspectral unmixing framework, enabling the transformation from noise to abundance maps. First, we embed the diffusion process into an autoencoder (AE) architecture and introduce a denoising guidance generator to extract robust guidance from hyperspectral images, directing the reverse diffusion process to produce spatially aligned and physically plausible abundances. Second, we design an abundance evaluation module that supervises the original data, that is, abundances in this context, using decoder-based reconstruction evaluation and unmixing prior knowledge. Additionally, we adopt a stage-wise freezing training strategy to progressively stabilize the model and address the learning optimization problem. Experimental results on both simulated and real-world datasets demonstrate that the proposed method effectively addresses the challenges of unsupervised hyperspectral unmixing, producing accurate, physically meaningful, and robust unmixing results.**

***Index Terms*— Autoencoder (AE) network, diffusion model, feature extraction, hyperspectral unmixing.**

I. Introduction

HYPERSPECTRAL imaging is an advanced technology that captures high-resolution spectral information, integrating spatial and spectral data to provide unique spectral signatures for each pixel. This capability allows for precise identification and differentiation of materials, making it a valuable tool in fields such as environmental management, precision agriculture, urban planning, and food safety [1], [2]. However, due to the low spatial resolution of hyperspectral imaging devices, the diversity and intimate interactions of materials in the imaged scenes, and multiple photon reflections from layered objects, a single pixel in a hyperspectral image often captures multiple distinct materials, leading to performance degradation in further analysis [3], [4]. To address this issue, hyperspectral unmixing has emerged as a crucial technique. This technique decomposes each pixel into pure spectral signatures, called endmembers, and their corresponding abundance fractions. The process of hyperspectral unmixing typically encompasses several key steps: mixture model assumption, endmember extraction, and abundance inversion [5]. By accurately identifying and quantifying these components, hyperspectral unmixing enhances the understanding and analysis of complex scenes, making it an essential tool in hyperspectral data processing.

Linear mixing models (LMMs) [6] is a widely utilized approach in hyperspectral unmixing, due to its simplicity and effectiveness in various applications. LMMs assume that surface materials are distributed in a mosaic form and that light interacts with a single material before being captured by remote sensing sensors. The unmixing task modeled by LMMs can be regarded as an inverse problem to restore endmembers and abundances. For instance, geometric methods such as the N-FINDR [7] and vertex component analysis (VCA) [8] are widely used for extracting endmembers from hyperspectral data. Fully constrained least-squares (FCLS) [9] are employed to estimate the abundance fractions. Unsupervised methods such as nonnegative matrix factorization (NMF) jointly estimate both endmembers and abundances directly from the observed data. To address the inherent nonconvexity of the NMF optimization problem and improve solution quality, various constraints and regularization techniques have been introduced. Nonlinear mixing models (NLMMs) address complex light-material interactions, such as multiple scattering or intimate mixing. For example, kernel NMF (KNMF) [10] maps hyperspectral data into a high-dimensional space to approximate nonlinear mixing, while the generalized bilinear model [11] incorporates bilinear interaction terms for greater flexibility. Nevertheless, these methods rely on prior knowledge about scenarios and result in difficulties in constraining choices.

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In recent years, data-driven methods [12], [13] based on deep learning have emerged as solutions to the unmixing problem, reducing the need for complex prior knowledge to some extent. Through data training, data-driven methods can learn the relationships between observed spectral data and their components directly from the data. Unsupervised hyperspectral unmixing can be viewed as a process of modality transformation among the original hyperspectral data, latent feature representations, and abundance maps. In this context, the autoencoder (AE) framework serves as an effective tool, where the encoder is responsible for feature extraction, capturing essential spectral and spatial information from the high-dimensional HSI data, and transforming these features into abundance representations that satisfy nonnegativity and sum-to-one constraints. The decoder reconstructs the original HSI to estimate endmembers and validate the accuracy of the unmixing process. For instance, Deshpande et al. [14] proposed a self-supervised two-stage AE-based framework to jointly estimate endmembers and abundances, demonstrating robustness under high noise. Similarly, Zouaoui et al. [15] emphasized AE flexibility in hyperspectral unmixing, highlighting improvements through spectral angle loss, sparsity constraints, and convolutional structures. Many researchers have incorporated convolutional neural networks (CNNs) into these methods to fully exploit both spectral and spatial information [16]. For example, CNNAEU [17] employed 2D-CNNs to leverage local spatial features of pixels and analyzed the impact of spatial neighborhoods of abundance maps on pixel spectral reconstruction. DeepTrans [18] introduced Transformers to capture nonlocal long-distance spatial dependencies for unmixing.

Due to their ability to directly model the probability distributions of data, generative methods, such as variational AEs (VAEs) [19] and generative adversarial networks (GANs) [20], are suitable for capturing complex interactions across modalities, effectively accomplishing unmixing as a complex mapping from high-dimensional spectral data to low-dimensional abundance representations. DeepGUn [21] proposed a deep generative endmember model based on a VAE to learn spectral variability directly from observed data in an unsupervised manner, applying it to spectral unmixing through an iterative nonlinear least-squares optimization. AAENet [22] proposed an adversarial AE network for unsupervised unmixing, utilizing superpixel-based representations to effectively address challenges in data reconstruction. However, VAEs often suffer from blurry reconstructions due to their reliance on a probabilistic latent space [23], which can compromise the quality of generated data. GANs, on the other hand, are prone to instability during training, with issues such as mode collapse and adversarial optimization difficulties [24], which make them challenging to train and less reliable in capturing complex data distributions.

To address these limitations, diffusion models provide an effective generative framework. Diffusion models learn data distributions by gradually adding noise to the data and then reversing this process through iterative denoising. They can be seen as a form of evolutionary algorithm, as the iterative denoising in diffusion models mirrors the selection and variation processes in evolution [25]. By modeling the stepwise transformation from noise to the original data, they effectively capture complex data structures and provide high-quality reconstructions. UnmixDiff [26] applies diffusion models to hyperspectral image synthesis, demonstrating their potential in hyperspectral tasks. DiffUn [27] is a diffusion-model-based approach that formulates the unmixing process as a sampling task from a posterior distribution. However, as a semi-supervised method, DiffUn relies on the availability of a spectral library, which may limit its applicability in scenarios where prior spectral information is incomplete, inaccurate, or unavailable.

The advantage of using diffusion models for hyperspectral unmixing lies in their flexibility and efficiency. Compared to traditional AE-based unmixing methods, which typically extract abundances from hyperspectral images in a single step, diffusion models achieve this task through a gradual denoising process, offering greater flexibility and control. In single-step methods, insufficient learning in certain network layers can cause errors to propagate to the output, compromising the final results. In contrast, the generation process of diffusion models can be seen as a form of progressive optimization, and each step's update direction aligns with the tangent space of the data manifold [28], which disperses the impact of errors, enhancing the robustness and accuracy of the unmixing process. Furthermore, while traditional AEs operate directly on high-dimensional hyperspectral images, diffusion models work in a low-dimensional space, transforming noise into abundances, which share the same dimensions. This low-dimensional approach simplifies network design, improves efficiency, and enables diffusion models to better capture the spatial and structural relationships in the data.

However, applying diffusion models to unsupervised unmixing tasks still presents several challenges. First, the generative process of diffusion models is inherently stochastic, which requires a large number of sample pairs consisting of original signals and their corresponding noise, and the use of guidance. However, in unsupervised hyperspectral unmixing tasks, the abundance vectors are often unknown, and the original hyperspectral image is typically noisy and unsuitable to serve directly as guidance. Second, the training of diffusion models traditionally focuses on aligning the predicted noise distribution with the added noise distribution. In the reverse process, while the model explicitly estimates the noise at each step, it also implicitly refines a coarse approximation of the final generated sample. Unlike general generative tasks, where it is often difficult to evaluate the quality of generated samples, hyperspectral unmixing benefits from rich prior knowledge about the unmixing task, allowing for more interpretable and domain-specific evaluation metrics. Third, the complexity of the network structure for unsupervised unmixing problems poses a significant challenge. These networks typically consist of both an encoder and a decoder, forming a structurally complex system that corresponds to a challenging learning optimization problem.

To address these issues, we propose a novel diffusion-based unsupervised hyperspectral unmixing. Unlike classical AE-based unmixing methods, which directly transform

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high-dimensional hyperspectral data into a low-dimensional abundance matrix, our method gradually evolves a noise matrix, which has the same dimensions as the abundance matrix, into the abundance matrix through multiple denoising steps guided by low-dimensional guidance and prior knowledge. To address the stochasticity of the generated abundances, we propose a denoising guidance generator that extracts meaningful features from the hyperspectral image and employs them as robust guidance to guide the diffusion process. This generator works alongside general hyperspectral unmixing priors to provide robust guidance, ensuring the generation of desired abundances. To avoid the need for sample pairs and to ensure the quality and reliability of the generated abundances during both training and inference, we design an abundance evaluation module. This module consists of three components: noise evaluation to ensure alignment between predicted and expected noise distributions, a knowledge base to incorporate prior domain knowledge, and reconstruction evaluation to verify the spatial consistency between the abundances and the image patches. To enhance the training process, we adopt a stage-wise freezing strategy, which improves the stability and efficiency of training, making the diffusion model more practical and scalable for real-world applications. The major contributions of this article are summarized as follows.

1) To the best of our knowledge, the proposed method is the first work to apply diffusion models to the unsupervised hyperspectral unmixing task. Under the guidance, the diffusion model is applied to achieve the transformation from noise to abundance maps.
2) A denoising guidance generator is proposed to extract features from hyperspectral images and generate denoising low-dimensional guidance, effectively guiding the diffusion process and reducing stochasticity in abundance generation. An abundance evaluation module that combines noise distribution, prior knowledge, and reconstruction error is introduced to guide model training and produce meaningful abundance maps.
3) A stage-wise freezing strategy is employed to handle the complexity of the network structure in unsupervised unmixing. This strategy dynamically adjusts the trainable parameters of the decoder during the training process, allowing the model to first stabilize the encoder and then jointly optimize the entire network.

The rest of this article is organized as follows. Section II briefly introduces the linear mixing model and related works about AE-based unmixing approaches and the conditional diffusion models. In Section III, the proposed method is presented in detail. The performance of the proposed method on synthetic and real hyperspectral datasets is evaluated in Section IV. Finally, comprehensive conclusions are drawn in Section V.

II. Related Works

In this section, we provide an introduction to the related work associated with our proposed method. Specifically, we introduce the linear mixing model as the foundational framework for hyperspectral unmixing, followed by a discussion of AE-based unmixing methods. Finally, we present the principles of conditional diffusion models, highlighting their potential for addressing challenges in hyperspectral unmixing.

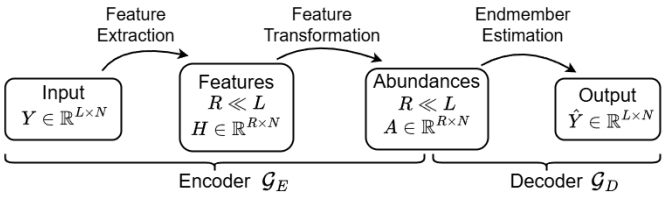


Fig. 1. Framework of AE-based unmixing.

A. Linear Mixing Model

The LMM is a widely used hyperspectral unmixing method, which assumes the HSI can be expressed as a linear combination of a few pure spectral features [29]. Let $\mathbf{Y} \in \mathbb{R}^{L\times N}$ denote the observed HSI matrix, where L represents the number of spectral bands and N represents the number of pixels. The LMM can be formulated as

$$\mathbf{Y} = \mathbf{MA} + \mathbf{N} \tag{1}$$

where $\mathbf{M} \in \mathbb{R}^{L\times M}$ is the endmember matrix containing M pure spectral signatures, $\mathbf{A} \in \mathbb{R}^{M\times N}$ is the abundance matrix representing the proportions of each endmember in the pixels, and $\mathbf{N} \in \mathbb{R}^{L\times N}$ is the noise matrix accounting for measurement errors. The primary objective of hyperspectral unmixing is to estimate $\mathbf{M}$ and $\mathbf{A}$, given $\mathbf{Y}$. The abundance matrix $\mathbf{A}$ is subject to two constraints [30]: the abundance nonnegativity constraint (ANC), $a_{ij} \geq 0 \forall i, j$, and the abundance sum-to-one constraint (ASC), $\sum_{i=1}^{M} a_{ij} = 1 \forall j$. These ensure physically meaningful and interpretable abundance estimations within the LMM.

B. AE-Based Unmixing

Unsupervised hyperspectral unmixing can be viewed as a process of modality transformation among the original hyperspectral data, latent feature representations, and abundance maps. AE-based unmixing methods naturally align with this perspective, as they leverage the architecture of an AE to perform such transformations in an unsupervised manner. Specifically, in AE-based unmixing, the process typically involves four main steps: feature extraction, feature transformation to abundance, endmember estimation, and abundance evaluation based on HSI reconstruction. The encoder is responsible for extracting latent features from the original hyperspectral data, which are then transformed into abundance maps that satisfy physical constraints. The decoder subsequently reconstructs the hyperspectral image using the estimated abundance maps and endmembers, ensuring consistency between the original and reconstructed data. Fig. 1 shows the framework of AE-based unmixing.

The first step in AE-based unmixing is feature extraction. The encoder processes the input hyperspectral data Y and extracts a compressed latent representation $\mathbf{H} \in \mathbb{R}^{R\times N}$. This representation reduces the dimensionality of the input data

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while preserving its most relevant spectral information. This step is critical because the effectiveness of feature extraction directly influences the accuracy of the subsequent unmixing process.

The second step is feature transformation, where the latent representation $\mathbf{H}$ is mapped to the abundance matrix $\mathbf{A} \in \mathbb{R}^{R\times N}$. This mapping ensures that the extracted features are aligned with the physical interpretation of abundances. The transformation incorporates prior knowledge, such as nonnegativity, the sum-to-one constraint, and sparsity, to ensure that the abundance matrix is physically meaningful.

In the final step, data reconstruction, the decoder reconstructs the original hyperspectral data $\hat{\boldsymbol{Y}} \in \mathbb{R}^{L\times N}$ using the estimated abundance matrix and the endmember matrix. To ensure that the reconstructed data $\hat{\boldsymbol{Y}}$ closely approximates the input data $\mathbf{Y}$, the reconstruction loss is minimized

$$\mathcal{L}_{\text{recon}} = \left\|\mathbf{Y} - \hat{\boldsymbol{Y}}\right\|^2. \tag{2}$$

Among the three steps, feature extraction and feature transformation are the most critical. Feature extraction determines how well the encoder captures the spectral and spatial characteristics of the data, while feature transformation ensures that these features are mapped into physically meaningful abundance representations.

C. Conditional Diffusion Models

Diffusion models have emerged as the new state-of-the-art family of deep generative models. The inspiration for diffusion models comes from nonequilibrium thermodynamics. A Markov chain of diffusion steps is defined, gradually adding random noise to the real data, and then learning the reverse diffusion process to construct the desired data samples from the noise. Initially, diffusion models were primarily applied in image synthesis tasks. Subsequently, due to their powerful data generation capabilities and ability to approximate complex data distributions, they have shown potential in various fields, such as computer vision [31], [32], natural language processing [33], [34], and multimodal modeling [35], [36].

In particular, denoising diffusion probabilistic model (DDPM) [37] is one of the most widely used diffusion models, which gradually transforms a Gaussian noise distribution x_t into a complex data distribution x_0 using a latent variable model consisting of two processes: the forward diffusion process and the reverse denoising process. To extend DDPM for tasks requiring specific outputs, such as controlled data generation in fields like text-to-image synthesis [38], text-to-audio generation [39], and video generation [40], conditional diffusion models introduce additional conditioning information y into the reverse process. y can represent labels, class embeddings, or other auxiliary information. By conditioning on y, the diffusion model generates samples that align with the desired attributes or constraints.

In the forward diffusion process, the data distribution x_0 is gradually transformed into a Gaussian noise distribution x_T through a sequence of intermediate latent variables $x_1, x_2, \ldots, x_T$. At each time step t, Gaussian noise is added to the data, and the process can be defined as

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t}\cdot x_{t-1}, \beta_t \cdot \mathbf{I}\right) \tag{3}$$

where β_t is a variance schedule that controls the amount of noise added at each step. This forward process ensures that x_T approaches an isotropic Gaussian distribution as $t \to T$.

To reverse this process, the reverse denoising process iteratively removes noise from x_t to recover the original data distribution x_0. This reverse process is parameterized by a neural network ϵ_θ, which predicts the noise component in x_t. The reverse process can be expressed as

$$p_\theta(x_{t-1} \mid x_t, y) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t, y), \Sigma_\theta(t)) \tag{4}$$

where $\mu_\theta(x_t, t, y)$ is the predicted mean, depending on x_t, the time step t, the conditioning information y, and $\Sigma_\theta(t)$ is the predicted variance.

The estimate of the original data $\hat{x}_0$ can be reconstructed from the noisy sample x_t and the predicted noise ϵ_t at a given diffusion step t

$$\hat{x}_0 = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \sqrt{1-\alpha_t}\epsilon_t\right) \tag{5}$$

where α_t is a time-dependent scaling factor and ϵ_t represents the predicted noise at timestep t. This formulation shows that the reverse process, while explicitly predicting noise ϵ_t, also implicitly refines $\hat{x}_0$ at each step, gradually reconstructing the original data distribution. At each step of the reverse process, $\hat{x}_0$ becomes progressively closer to the original data.

The training objective of the diffusion model is designed to minimize the difference between the true noise ϵ added during the forward process and the noise predicted by the neural network $\hat{\epsilon}_\theta$, which is defined as

$$\mathcal{L} = \mathbb{E}_{x_0,\epsilon,t,y}\left[\left\|\epsilon - \hat{\epsilon}_\theta(x_t, t|y)\right\|^2\right] \tag{6}$$

where x_t is a noisy sample generated at time step t, ϵ is the ground-truth noise added to the data, and y represents optional conditioning information, which is usually relevant information about the samples to be generated, including features and priors.

In the field of sample generation, traditional diffusion models typically transform random noise into samples with certain physical significance through a progressive refinement process. However, due to the lack of relevant information, these models often evaluate only the estimated noise during the transformation. For specific applications, such as image restoration and super-resolution, and especially hyperspectral unmixing, partial information related to the desired generated samples, as well as general prior knowledge, is usually available. This information can be effectively utilized to guide the transformation process, thereby improving the quality and reliability of the generated results.

III. Proposed Methods

In this section, we introduce the generative framework for unsupervised hyperspectral unmixing with diffusion models in

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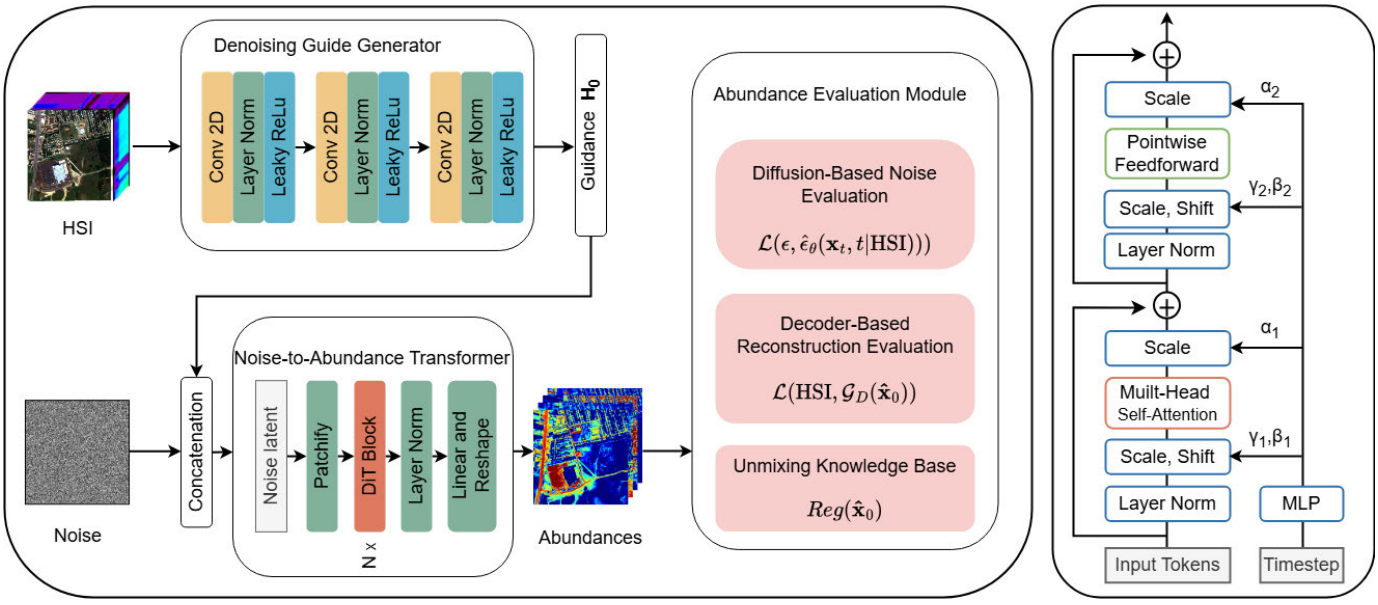


Fig. 2. Framework of the proposed method. Hyperspectral image patches are first processed by the denoising guidance generator to produce robust low-dimensional guidance, followed by a noise-to-abundance Transformer using diffusion models to map noise to abundance maps, and finally, the unmixing results are evaluated for quality and reliability through the abundance evaluation module. The right image presents the detailed structure of the Dit layer.

detail. Under the guidance of the unique characteristic information and general features of hyperspectral images, noise is transformed into an abundance matrix block corresponding to a specific hyperspectral image block. Its framework is shown in Fig. 2. The network comprises three main modules: a denoising guidance generator, a noise-to-abundance Transformer, and an abundance evaluation module. Hyperspectral image patches are first processed by the denoising guidance generator, which consists of three consecutive 2-D convolutional layers that progressively extract and compress spatial–spectral features to generate low-dimensional guidance. The guidance integrates spectral characteristics and spatial contextual information from the hyperspectral patches, injected into the diffusion model of the noise-to-abundance Transformer through channel concatenation to constrain the generation process. The noise-to-abundance Transformer processes the noisy input by linearly projecting it into vector sequences and iteratively refining the features through DiT blocks, ultimately performing the transformation from Gaussian noise to physically interpretable abundance maps. Finally, the generated abundance maps are evaluated through the abundance evaluation module based on the characteristics of hyperspectral image blocks and general prior knowledge to ensure their quality and reliability. The abundance evaluation module primarily consists of three components: the noise evaluation based on the diffusion process monitors reverse diffusion convergence by comparing predicted and theoretical noise distributions, the decoder reconstruction evaluation calculates the spectral mean squared error between original and reconstructed hyperspectral patches, and the unmixing knowledge base enforces prior constraints. Given the complexity and high stochasticity of training diffusion models, we adopt a stage-wise freezing strategy for the decoder to accelerate the training process—initially freezing the decoder to prioritize optimization of the diffusion branch before gradually unfreezing parameters in later stages. A detailed description of the modules is provided as follows.

A. Denoising Guidance Generator

The denoising guidance generator is a critical component of the proposed unsupervised hyperspectral unmixing framework. The output guidance information consists of two parts: one is the information related to the desired generated samples, and the other is the general prior knowledge. Its primary function is to guide the diffusion process by providing meaningful constraints on the generated abundance maps, thereby addressing the inherent stochasticity of the diffusion model. Specifically, the denoising guidance generator transforms noisy hyperspectral images into denoised, low-dimensional guidance. This not only enhances the robustness of the framework but also facilitates the training of the diffusion model by supplying clean and structured guidance. Moreover, by reducing the dimensionality, the denoising guidance generator lowers the computational cost of the diffusion model.

The original hyperspectral image patch, denoted as $\mathbf{Y}_{\text{patch}} \in \mathbb{R}^{L\times k\times k}$, is a small region of the larger hyperspectral image, containing localized spectral–spatial information, where k is the patch size. It is fed into the denoising guidance generator, which is composed of three layers: a 2-D convolutional layer captures spatial hierarchies and patterns present within the patch; a normalization layer ensures stable training and improves convergence by normalizing feature maps; and a LeakyReLU activation function introduces nonlinearity to model complex feature interactions while preventing dead neurons. These layers work together to transform the hyperspectral image patch $\mathbf{Y}_{\text{patch}}$ into latent variables, denoted as $\mathbf{H}_0$, which are rich in semantic content and represent the patch-level features.

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The process can be mathematically expressed as follows:

$$\mathbf{H}_0 = f_{\text{CNN}}\left(\mathbf{Y}_{\text{patch}}\right) \tag{7}$$

where f_{CNN} represents the transformation applied by the denoising guidance generator, encapsulating convolution, normalization, and activation operations.

The derived $\mathbf{H}_0$ provides crucial spectral–spatial information of the hyperspectral image in a compact, low-dimensional representation. Acting as conditional guidance, $\mathbf{H}_0$ is concatenated with the input noise along the channel dimension to form the input to the diffusion model, which enables the model to effectively incorporate the spectral characteristics captured in $\mathbf{H}_0$ while maintaining spatial coherence throughout the denoising process.

B. Noise-to-Abundance Transformer

The noise-to-abundance Transformer in our network framework converts noise into abundances, where the noise matrix and the abundance matrix share the same dimensions. This transformation is achieved through an iterative process under guidance. In this framework, noise is assumed to be generated by progressively adding controlled noise to the original data $\mathbf{x}_0$, resulting in a noisy sample $\mathbf{x}_T$ at the final timestep T. This process follows the forward diffusion process in diffusion models, where the data becomes increasingly corrupted at each step. The step-by-step noise addition process is mathematically defined as

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1}, \quad t \in \{0, 1, \ldots, T\} \tag{8}$$

where $\mathbf{x}_t$ represents the data at timestep t, α_t is the noise scheduling coefficient at timestep t, and $\boldsymbol{\epsilon}_{t-1}$ denotes Gaussian noise introduced at the previous timestep.

By drawing on the evolutionary interpretation of diffusion models, the reverse diffusion process can be seen as an iterative evolutionary refinement step [25], where the model, guided by fitness based on performance evaluation, progressively reconstructs a more optimal solution. In this context, each denoising step functions analogously to a selection event in evolutionary processes, wherein less optimal variations are gradually eliminated, allowing the underlying structure of the data to emerge more clearly.

Mathematically, this process aligns with the principle of survival of the fittest, where each timestep t refines the abundance matrix $\mathbf{x}_t$ by selectively removing noise components that deviate from the learned spectral characteristics. The noise predictor $\hat{\boldsymbol{\epsilon}}_\theta$ acts as a learned selection mechanism, analogous to adaptive fitness landscapes in evolutionary algorithms, where solutions are iteratively improved by filtering out less favorable perturbations. At each step, the model updates the abundance estimate using a learned gradient, akin to a selection gradient in evolutionary optimization

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\hat{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t|\mathbf{Y}\text{patch})\right) + \sigma_t \mathbf{z}. \tag{9}$$

From an optimization perspective, this process resembles an adaptive evolutionary strategy, where the network refines the solution through a form of stochastic gradient descent over multiple iterations. The introduction of noise at each step can be viewed as an exploratory mutation, ensuring the model does not become trapped in local optima, while the denoising process acts as a form of selective pressure, gradually steering the solution toward a globally coherent and spectrally consistent abundance representation.

This reverse process is parameterized as shown in (4), which predicts the noise component in $\mathbf{x}_t$. To effectively utilize hyperspectral image patches $\mathbf{Y}$patch, the reverse diffusion process is reformulated as a maximization problem. The goal is to select a sample based on the current information such that the following probability, which represents the performance evaluation or fitness, is maximized [41]:

$$\begin{aligned}\log p_\theta\left(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{Y}_{\text{patch}}\right) &= \log\left(p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \cdot p\left(\mathbf{Y}_{\text{patch}} \mid \mathbf{x}_t\right)\right) \\ &\quad + C_1 \\ &\approx \log(\mathcal{Z}) + C_2\end{aligned} \tag{10}$$

where C_1 and C_2 are constants. $\mathcal{Z} \sim \mathcal{N}(\mu_\theta(\mathbf{x}_t, t) + \mathbf{G}, \Sigma_\theta)$ and $\mathbf{G} = \nabla_{\mathbf{x}_t} \log p(\mathbf{Y}\text{patch} \mid \mathbf{x}_t)$ represent the gradient guidance information. The performance depends on two aspects: the alignment of the estimated noise with the actual noise distribution and the evaluation of $\hat{\mathbf{x}}_0$ based on the guidance information.

The estimate of the original abundance matrix $\hat{\mathbf{x}}_0$ can be reconstructed from the noisy sample $\mathbf{x}_t$ and the predicted noise at a given diffusion step as shown in (5). Thus, $\mathbf{G} \approx \nabla_{\mathbf{x}_t} \log p(\mathbf{Y}\text{patch} \mid \hat{\mathbf{x}}_0)$. We define a prior function $p(\mathbf{Y}\text{patch} \mid \hat{\mathbf{x}}_0)$ that represents the relationship between hyperspectral image patches and the abundance matrix [42]

$$p\left(\mathbf{Y}_{\text{patch}} \mid \hat{\mathbf{x}}_0\right) = \exp\left[-\mathcal{Q}\left(\hat{\mathbf{x}}_0\right)\right] \tag{11}$$

where the guidance term $\mathcal{Q}(\hat{\mathbf{x}}_0)$ integrates the hyperspectral observation data with existing constraints

$$\mathcal{Q}\left(\hat{\mathbf{x}}_0\right) = \lambda \cdot \mathcal{L}\left(\hat{\mathbf{x}}_0, \mathbf{Y}\text{patch}\right) + \mu \cdot \text{Reg}\left(\hat{\mathbf{x}}_0\right). \tag{12}$$

Here, $\mathcal{L}(\hat{\mathbf{x}}_0, \mathbf{Y}\text{patch}) = \|\mathbf{Y}_{\text{patch}} - \mathcal{G}_D(\hat{x}_0)\|_2^2$ is the reconstruction error in the AE framework to evaluate unmixing performance, $\text{Reg}(\hat{\mathbf{x}}_0)$ is the regularization term that incorporates general prior knowledge for unmixing, and λ, μ are weighting parameters.

The gradient guidance term $\mathbf{G}$ can be expressed as

$$\mathbf{G} = \nabla_{\mathbf{x}t} \log p\left(\mathbf{Y}\text{patch} \mid \hat{\mathbf{x}}_0\right) = -\nabla \mathbf{x}_t \mathcal{Q}\left(\hat{\mathbf{x}}_0\right). \tag{13}$$

Through this gradient guidance mechanism, our diffusion model can consider hyperspectral observation data and prior constraints during the reverse process, generating more accurate and physically interpretable abundance matrices. In each step of the reverse diffusion process, the model optimizes abundance estimates using predicted noise and gradient guidance, ultimately transforming initial noise $\mathbf{x}_T$ into physically meaningful abundance results $\hat{\mathbf{x}}_0$.

As the core predictive model $\hat{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t|\mathbf{Y}\text{patch})$, the diffusion Transformer (DiT) [43] combines the diffusion process with the representational power of Transformer [44] architectures, outperforming the commonly used U-Net backbone in diffusion models. The DiT block processes latent patches by first applying adaptive layer normalization conditioned on the

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timestep, followed by multihead self-attention and a feed-forward network with a specific zero initialization for the final layer. This design allows the model to effectively capture both temporal and spatial dependencies in the latent space, enabling it to learn complex relationships across timesteps and conditions. By leveraging the self-attention mechanism, the DiT efficiently models the global context of the abundance matrix $\mathbf{x}_t$, while integrating $\mathbf{H}_0$ to ensure spectral consistency. This synergy between the diffusion process and the Transformer architecture enhances the model's ability to generate accurate and robust abundance estimates, improving performance in hyperspectral unmixing tasks.

Through the integration of the noise-to-abundance Transformer, our framework leverages the strengths of diffusion models, such as their flexibility and low-dimensional processing, to enhance both efficiency and output quality, providing a compelling alternative to traditional AE-based approaches.

C. Abundance Evaluation Module

The estimation problem of $\mathbf{x}_{t-1}$ can be reformulated as searching for a $\mathbf{x}_{t-1}$ that maximizes the probability $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$. The abundance evaluation module is used to assess the performance of $\mathbf{x}_{t-1}$. Traditional diffusion models primarily focus on minimizing noise prediction errors, with limited attention given to supervising or guiding the generation of $\mathbf{x}_{t-1}$. In unsupervised unmixing tasks, the generated abundances are informed by prior knowledge and are required to correspond spatially to the image patches. To address this limitation and effectively incorporate unmixing prior knowledge, we propose an abundance evaluation module. Specifically, this module maximizes (10) by maximizing two key components: the conditional probability term $\log(p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t))$, which requires the predicted noise to match the statistical distribution of the added noise, and the prior probability term $\log(p(\mathbf{Y}_{\text{patch}} \mid \mathbf{x}_t))$, which requires minimizing (12) to reduce both the reconstruction error and the sparsity constraint term. To meet these objectives, this module comprises three components: diffusion-based noise evaluation, which ensures alignment between the predicted and added noise distributions; the incorporation of unmixing prior knowledge, which enforces sparsity constraints on the abundance maps; and decoder-based reconstruction evaluation, which employs a decoder to supervise and verify the spatial consistency between the abundances and the image patches. Each component is described in detail below.

The diffusion model component of the loss is responsible for learning the underlying data distribution by denoising the input at each diffusion step. Specifically, the model predicts the added noise in the diffusion process, and the noise prediction error is minimized as

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\mathbf{H}_0,\epsilon,t,\mathbf{Y}\text{patch}}\left[\left\|\epsilon - \hat{\epsilon}_\theta(\mathbf{x}_t, t|\mathbf{Y}\text{patch})\right\|^2\right] \tag{14}$$

where $\mathbf{H}_0$ is the guidance information, which is channel-wise concatenated with the noisy input. This loss term ensures that the diffusion model learns meaningful abundance representations by progressively denoising latent variables.

Hyperspectral pixels in unmixing tasks exhibit several key characteristics. Neighboring pixels within each image patch are typically similar and consistent, resulting in spatial sparsity and structural sparsity. Spatial sparsity assumes that only a subset of endmembers is active within an image patch, leading to a sparse abundance matrix. Additionally, pixels within the same image patch are typically correlated, sharing the same endmembers in varying proportions. Therefore, we introduce a regularization term to constrain the solution, which is defined as

$$\text{Reg}(\hat{x}_0) = \|\hat{x}_0\|_{1/2} + \text{Tr}\left(\sqrt{\hat{x}_0^\top \hat{x}_0}\right) \tag{15}$$

where the first term, $\|\hat{x}_0\|_{1/2}$, promotes sparsity in the solution, and the second term, $\text{Tr}((\hat{x}_0^\top \hat{x}_0)^{1/2})$, encourages structural correlation by leveraging the low-rank property of the matrix $\hat{x}_0$. $\text{Tr}(\cdot)$ is the trace operator.

Decoder-based reconstruction evaluation is designed to assess the quality of the generated abundance maps by reconstructing the hyperspectral image and calculating the reconstruction loss. Specifically, the generated abundance maps are input into a decoder to produce the reconstructed hyperspectral image. By minimizing the reconstruction loss, the quality of the abundance maps can be indirectly evaluated. The reconstruction loss is defined as the squared Euclidean norm of the difference between the observed hyperspectral data and the reconstructed data

$$\mathcal{L}_{\text{recon}} = \left\|\mathbf{Y}_{\text{patch}} - \hat{\mathbf{Y}}_{\text{patch}}\right\|_2^2 = \left\|\mathbf{Y}_{\text{patch}} - \mathcal{G}_D(\hat{x}_0)\right\|_2^2 \tag{16}$$

where $\mathbf{Y}_{\text{patch}}$ is the observed hyperspectral data, $\hat{\mathbf{Y}}_{\text{patch}}$ is the reconstructed hyperspectral data, and $\mathcal{G}_D(\hat{x}_0)$ represents the decoder function that maps the estimated abundance matrix $\hat{x}_0$ back to the spectral domain.

Depending on the mixing model, the decoder can be implemented as either a linear decoder or a nonlinear decoder. The linear decoder is implemented as a linear fully connected layer without bias. This design assumes that the observed spectral signature of each pixel is a linear combination of endmember signatures, weighted by their respective abundances. In this article, due to the interpretability and characteristics of the dataset, we adopt a linear decoder while the proposed method can also be extended by augmenting LMM with an additive nonlinear fluctuation term to handle nonlinear or more complex real-world scenarios.

The total loss function is a weighted combination of these three terms and the process of solving this optimization problem is equivalent to achieving the unmixing objective

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\text{diff}}\left(\epsilon, \hat{\epsilon}_\theta\left(\mathbf{x}_t, t|\mathbf{Y}_{\text{patch}}\right)\right) + \lambda\text{Reg}(\hat{x}_0) \\ + \mu\mathcal{L}_{\text{recon}}\left(\mathbf{Y}_{\text{patch}}, \mathcal{G}_D(\hat{x}_0)\right)\end{aligned} \tag{17}$$

where λ controls the weight of the abundance regularization term and μ determines the relative importance of the reconstruction loss. This unified loss framework ensures that the model learns robust abundance representations, reconstructs hyperspectral images accurately, and produces sparse, physically meaningful abundance outputs across all phases of training.

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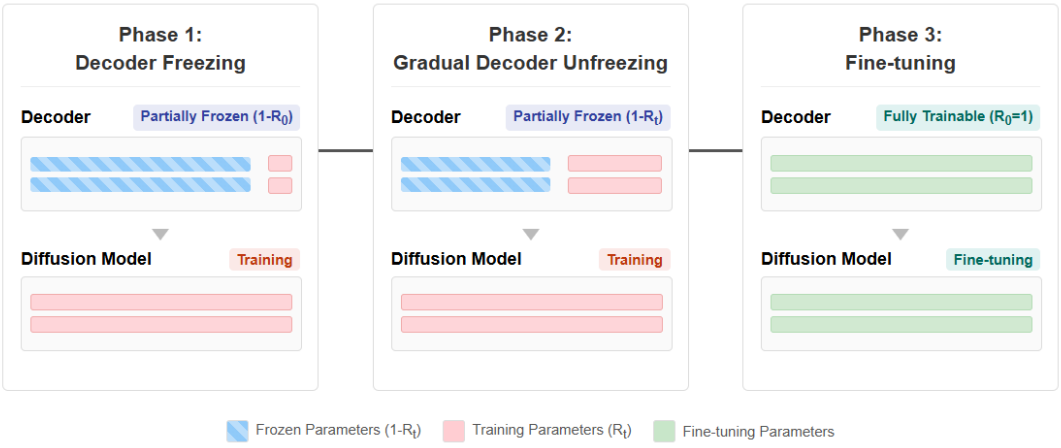


Fig. 3. Flowchart of the stage-wise freezing strategy.

D. Stage-Wise Freezing Strategy

The stage-wise freezing strategy introduces a novel training methodology designed to enhance the performance and stability of diffusion models, particularly when applied to abundance estimation. This approach addresses the challenge of training diffusion models effectively in the early stages, where model instability can hinder convergence and lead to suboptimal results. The flowchart of this strategy is shown in Fig. 3.

We utilize endmembers extracted via VCA [8] to initialize the decoder weights. A stage-wise freezing and unfreezing strategy, governed by the dynamically adjusted hyperparameter R_0, is applied to the decoder weights to enhance training stability and efficiency. R_0 represents the proportion of trainable decoder weights during a given training stage. It is initialized to a low value and progressively increases, incorporating more decoder weights into the optimization process. This progression is controlled by a decay factor, α, following an exponential decay pattern. The training process alternates between freezing and unfreezing phases, each lasting for a defined cycle length, T_c, balancing focused encoder training and joint network optimization. The training phases are as follows.

1) Decoder Freezing for Abundance Diffusion Model Training: In the initial training phase, the decoder weights are largely frozen by setting R_0 to a low value. This allows the diffusion model parameters to focus on learning the underlying abundance distribution without interference from decoder updates. The frozen decoder acts as a stabilizing factor, ensuring that the abundance representation is effectively captured during this stage.

2) Gradual Decoder Unfreezing for Joint Optimization: As training progresses, R_0 is gradually increased according to the following schedule:

$$R_t = R_0 \cdot \alpha^t \tag{18}$$

where t represents the cycle index and α is a decay factor. This dynamic adjustment progressively incorporates more decoder weights into the optimization process. Training alternates between phases of partially frozen (R_t) and fully unfrozen decoder states, each lasting T_c epochs. This strategy enables the decoder to adapt incrementally to the learned abundance representation, improving the reconstruction quality while maintaining training stability.

3) Fine-Tuning for Optimal Performance: In the final phase, R_0 reaches its maximum value of 1, allowing all decoder weights to be fully trainable. At this stage, the model is fine-tuned to achieve optimal performance. Regularization techniques are employed to prevent overfitting and ensure robust parameter learning.

IV. Experiments

In this section, we conduct extensive experiments to demonstrate the performance of the proposed unmixing model. Specifically, we first present the experimental settings. Then, the experimental results are reported and analyzed from different aspects.

A. Data Description

1) Synthetic Dataset: For the synthetic data experiments, five endmembers are randomly selected from the ASTER spectral library [45], that is, limestone, basalt, concrete, conifer, and asphalt. We choose one material as background, and the other four materials are randomly placed in the corner, whose shape, width, and location are sampled from Gaussian distributions [46]. The synthetic dataset consists of 60 × 60 pixels and 200 bands covering the wavelength range from 0.4 to 14 μm. Fig. 4(a) illustrates the pseudo-color image of the Synthetic dataset.

2) Samson Dataset: The original Samson dataset [47] contains 952 × 952 pixels and each pixel contains 156 spectral bands with wavelength ranging from 401 to 889 nm. A region containing 95 × 95 pixels is used to reduce computational cost. There are mainly three endmembers, that is, soil, tree, and water, in this area. Fig. 4(b) illustrates the pseudo-color image of the Samson dataset.

3) Jasper Ridge Dataset: The original Jasper Ridge dataset [48] has spatial dimensions of 512 × 614, covering a spectral range of 380–2500 nm with 224 bands. A 100 × 100

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subimage is utilized in this work. After removing the channels (1–3, 108–112, 154–166, and 220–224) vulnerable to water vapor and atmospheric conditions, 198 channels were finally retained for unmixing. There are mainly four endmembers, that is, trees, soil, water, and roads, in this area. Fig. 4(c) illustrates the pseudo-color image of the Jasper Ridge dataset.

4) Urban Dataset: The spatial dimension of the Urban dataset [49] is 307 × 307 and each pixel contains 210 spectral bands with the wavelength ranging from 400 to 2500 nm. After removing some water absorption bands (1–4, 76, 87, 101–111, 136–153, and 198–210), 162 bands describing the spectrum of each pixel are left for subsequent tests. There are mainly four endmembers, that is, asphalt, grass, trees, and roofs, in this area. Fig. 4(d) illustrates the pseudo-color image of the Urban dataset.

5) Cuprite Dataset: The Cuprite dataset [48] is a widely used hyperspectral dataset for unmixing tasks. It has a spatial size of 250 × 190 pixels and consists of 224 spectral bands in the wavelength range of 400–2500 nm. However, certain bands affected by the water absorption or low signal-to-noise ratio (SNR), such as bands 1–2, 104–113, and 148–167, are typically removed, leaving 188 valid bands for analysis. The dataset contains twelve well-known geological minerals: Alunite, Andradite, Buddingtonite, Chalcedony, Dumortierite, Kaolinite1, Kaolinite2, Muscovite, Montmorillonite, Nontronite, Pyrope, and Sphene. Fig. 4(e) illustrates the pseudo-color image of the Cuprite dataset.

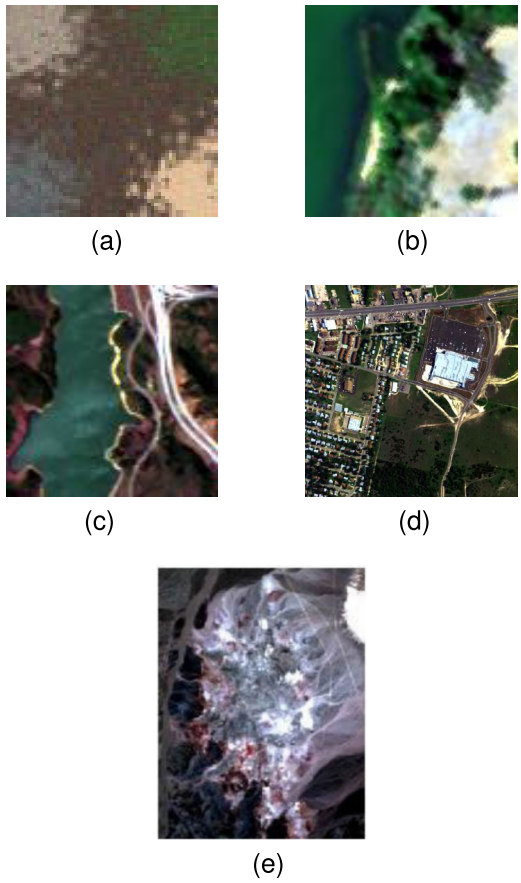


Fig. 4. Unmixing dataset. (a) Synthetic. (b) Samson. (c) Jasper Ridge. (d) Urban. (e) Cuprite.

B. Quantitative Metric for Unmixing Evaluation

Three evaluation indicators are used to assess the accuracy of the experimental results. The abundance estimation is assessed by the root-mean-square error (RMSE). The estimation performance in terms of endmembers is evaluated by computing the mean spectral angle distance (mSAD) [50]. Finally, the reconstructed error (RE) is considered to measure data fitting. They are defined as

$$\mathrm{RMSE} = \left(\frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{a}_i - \hat{\mathbf{a}}_i \right\|_2^2 \right)^{1/2} \tag{19}$$

$$\mathrm{SAD} = \sum_{i=1}^{M} \arccos\left(\frac{\mathbf{m}_i^{\top} \hat{\mathbf{m}}_i}{\|\mathbf{m}_i\| \cdot \left\|\hat{\mathbf{m}}_i\right\|} \right) \tag{20}$$

$$\mathrm{RE} = \frac{1}{\mathrm{NL}} \left\| \mathbf{Y} - \hat{\mathbf{Y}} \right\|_F^2 \tag{21}$$

where $\hat{\mathbf{a}}_i$ and $\mathbf{a}_i$ denote the ith estimated abundance vector and the ground-truth abundance related to it; $\hat{\mathbf{m}}_i$ and $\mathbf{m}_i$ mean the ith estimated endmember vector and the corresponding truth endmember, respectively. $\hat{\mathbf{Y}}$ are the reconstructed pixels.

C. Compared Algorithms

The performance of the proposed model is evaluated and compared to six different unmixing techniques. NMF-QMV [51] is a representative example of the classic NMF methods. Representing convolutional AE-based methods, CNNAEU [17], DeepTrans [18], and TANet [52] exploit spatial–spectral information for enhanced unmixing. Furthermore, DeepGUn [21] and AAENet [22] are representative unmixing methods based on VAEs and GANs, respectively. Finally, DiffUn [27] is the most recent and highly relevant unmixing method utilizing diffusion models.

1) *NMF-QMV:* This method proposes an automatic regularization parameter selection framework for hyperspectral unmixing. It systematically identifies the transition point where the simplex optimally encloses the data cloud, eliminating the need for manual tuning.
2) *CNNAEU:* This method introduces a spectral–spatial linear mixture model using a CNN AE to preserve spatial structure and estimate endmember and abundance maps directly from hyperspectral image patches.
3) *DeepTrans:* This method leverages Transformers for hyperspectral unmixing by combining a convolutional AE with a Transformer network, enhancing endmember spectra and abundance maps through long-range dependency capture.
4) *TANet:* TANet is a two-stream network for hyperspectral unmixing. It uses superpixel segmentation to extract endmember bundles, with one stream mapping pseudopure pixels to abundances and the other minimizing reconstruction errors through an untied-weighted AE.
5) *DeepGUn:* This method learns a deep generative EM model estimated with a VAE. Training with pure pixel information from the observed image enables an unsupervised approach to spectral unmixing, solving an alternating nonlinear least-squares problem iteratively

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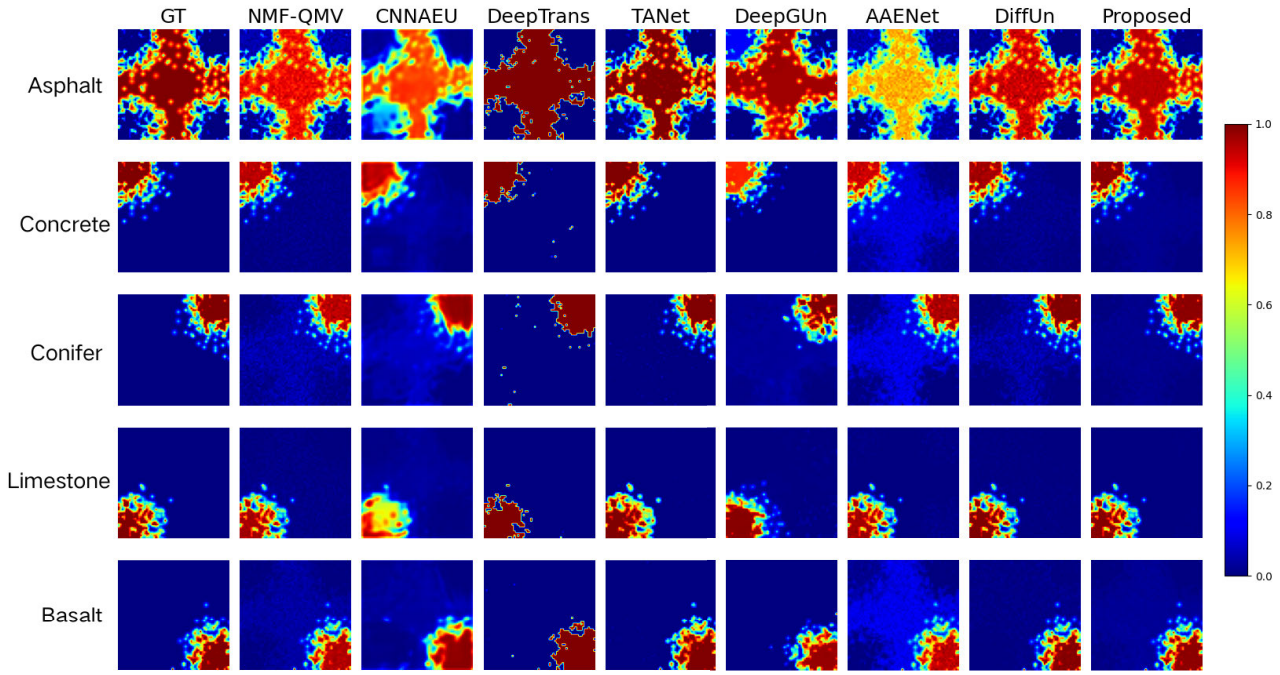


Fig. 5. Synthetic dataset SNR 20 dB: visual comparison of the abundance maps obtained by the different unmixing techniques.

for abundances and low-dimensional representations. This method also accounts for the spectral variability problem.

6) *AAENet:* AAENet is an unsupervised unmixing method based on adversarial AEs that partition images into homogeneous regions and employs adversarial training to transfer spatial information, ensuring pixels within the same region share statistical properties and follow specific prior distributions.
7) *DiffUn:* DiffUn is a semi-supervised unmixing method that uses diffusion models to sample abundances and endmembers from a posterior distribution. The prior is learned from a spectral library via an unconditional diffusion model, while the likelihood is derived from the observed data using the linear spectral mixture model.

D. Experimental Setup

NMF-QMV is run on the MATLAB R2021a platform. The proposed method is implemented using PyTorch and trained on an NVIDIA GeForce RTX 3090 GPU. For a fair comparison, in the following experiments, both the endmembers take VCA methods for initialization. The parameters for the comparison methods are set according to the configurations reported in their respective references. For the proposed method, key hyperparameters include the number of epochs to ensure convergence, patch size for defining the spatial context, and regularization coefficients λ, μ to balance diffusion noise error, reconstruction accuracy, and sparsity. The initial freezing ratio R_0 controlled the proportion of frozen decoder weights at the start of training, with a decay factor α progressively unfreezing them, and the cycle length T_c determining update frequency. Regarding the model parameters of the diffusion model, the time step T is set to 1000. The model was optimized using the Adam optimizer with a learning rate of 3×10^{-2}, ensuring stable and efficient training across diverse datasets. Detailed settings for each dataset are provided in Table I. The above parameters are empirical values derived from extensive experimentation conducted in this study. Similar to most current unmixing methods, the number of endmembers is assumed to be known for all datasets. Alternatively, it can also be determined using principal component analysis (PCA) [53] or virtual dimensionality (VD) [54], [55] by identifying the number of components that capture the majority of the data variance or signal dimensions.

TABLE I
HYPERPARAMETERS USED FOR TRAINING THE PROPOSED MODEL

Hyperparameters	Synthetic	Samson	Jasper	Urban
Epoch	600	1000	1000	1500
Patchsize	5	7	7	7
λ	1×10^{-2}	4×10^{-2}	8×10^{-2}	1×10^{-2}
μ	5	5	5	10
Learning rate	3×10^{-2}	3×10^{-2}	3×10^{-2}	3×10^{-2}
R_0	0.1	0.1	0.1	0.1
α	0.9	0.9	0.9	0.9
T_c	50	50	50	80

E. Experiments on Synthetic Data

To compare and analyze the robustness of the proposed method to noise, the additive noises with progressively decreasing SNRs of 30, 20, and 10 dB are added to generate the mixed pixels, which are sampled from the Gaussian distribution $\mathcal{N}(\mathbf{n}_n \mid \mathbf{0}, \mathbf{D})$ ($\mathbf{D} = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_R^2) \in \mathbb{R}^{R \times R}$). Table II presents aRMSE, mSAD, and RE results for

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TABLE II
QUANTITATIVE COMPARISONS ON THE SYNTHETIC DATASET WITH GAUSSIAN NOISE FROM SNR = 10-30 dB

		NMF-QMV	CNNAEU	DeepTrans	TANet	DeepGUn	AAENet	DiffUn	Proposed
aRMSE	10dB	0.1275	0.1755	0.1467	0.0824	0.1533	0.1921	0.0756	**0.0376**
	20dB	0.0396	0.1321	0.1462	0.0593	0.1500	0.1033	0.0258	**0.0256**
	30dB	0.0116	0.1223	0.1214	0.0575	0.1465	0.0226	0.0106	**0.0082**
mSAD	10dB	0.1560	0.1627	0.0635	0.0178	0.0783	0.1720	0.0849	**0.0176**
	20dB	0.0213	0.1352	0.0634	0.0102	0.0625	0.0990	0.0694	**0.0092**
	30dB	0.0066	0.1264	0.0519	0.0072	0.0621	0.0312	0.0684	**0.0052**
RE(e^{-2})	10dB	0.1085	0.4839	0.1318	0.4726	0.3018	0.5362	0.1138	**0.1011**
	20dB	0.0109	0.4160	0.0308	0.2806	0.1945	0.4744	0.0155	**0.0105**
	30dB	0.0013	0.3518	0.0154	0.2531	0.1949	0.3770	0.0056	**0.0010**

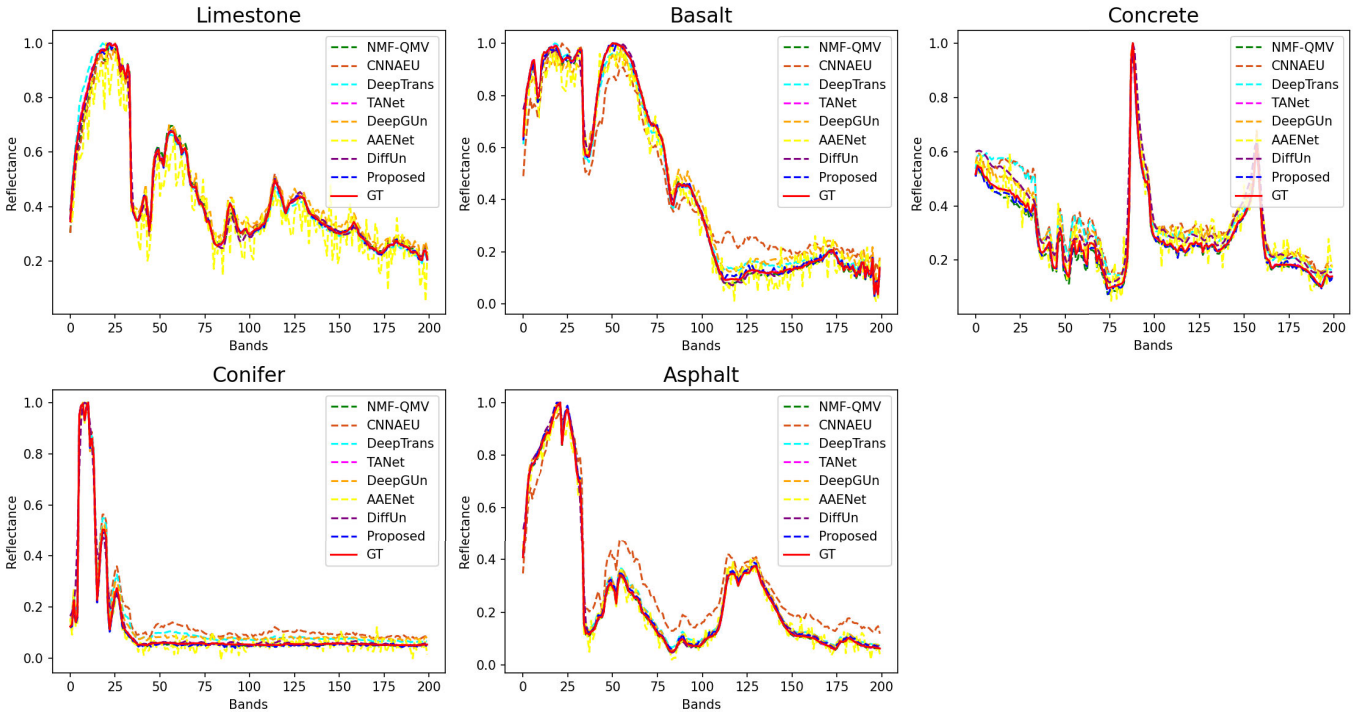


Fig. 6. Synthetic dataset SNR 20 dB: visual comparison of the endmembers obtained by the different unmixing techniques.

different unmixing methods on the Synthetic dataset, with the best results highlighted in bold. Note that all the algorithms are performed ten times on each dataset, and the average value is recorded for comparison. Overall, the performance of unmixing methods exhibits a certain degree of decline as the noise level increases. Under low-noise conditions, NMF-QMV and the proposed method demonstrate excellent unmixing performance. As the noise level increases, such as from 20 to 10 dB, the unmixing performance of NMF-QMV and AAENet is significantly affected, whereas our method demonstrates much better robustness to noise. In synthetic datasets, the performance of deep learning methods is often inferior to that of NMF or other traditional methods. This is typical because synthetic datasets are usually small in size and lack the diversity needed to effectively train deep learning models, which require large amounts of diverse data to generalize well. However, our method is still able to achieve good unmixing performance on such small datasets, due to its ability to effectively capture the underlying spectral features and its robustness to limited data sizes.

The estimated abundance maps when the SNR is set at 20 dB are shown in Fig. 5. Fig. 6 illustrates the estimated

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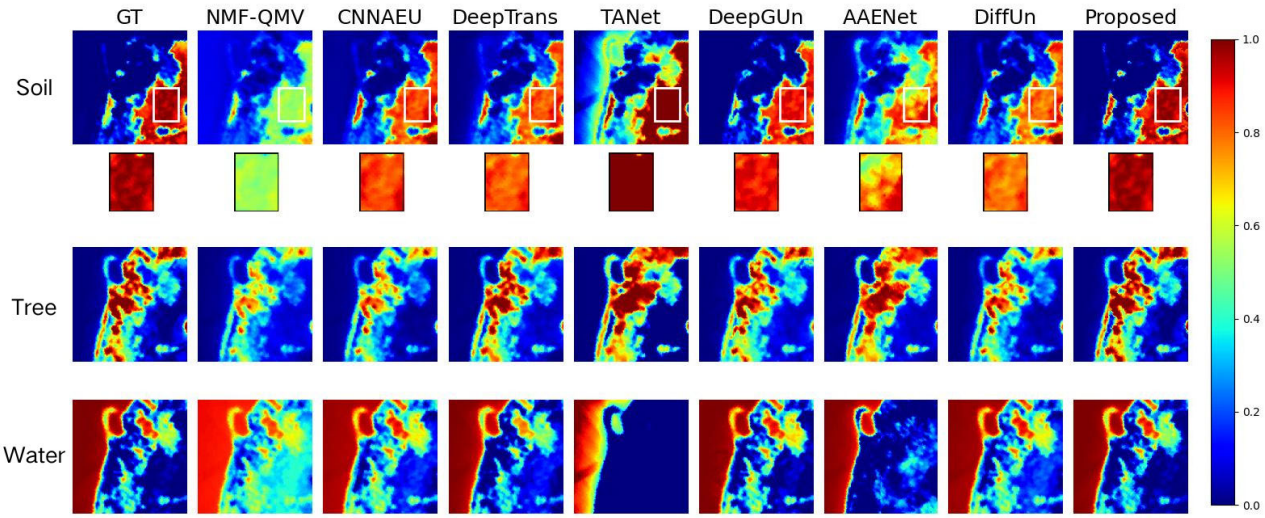


Fig. 7. Samson dataset: visual comparison of the abundance maps obtained by the different unmixing techniques.

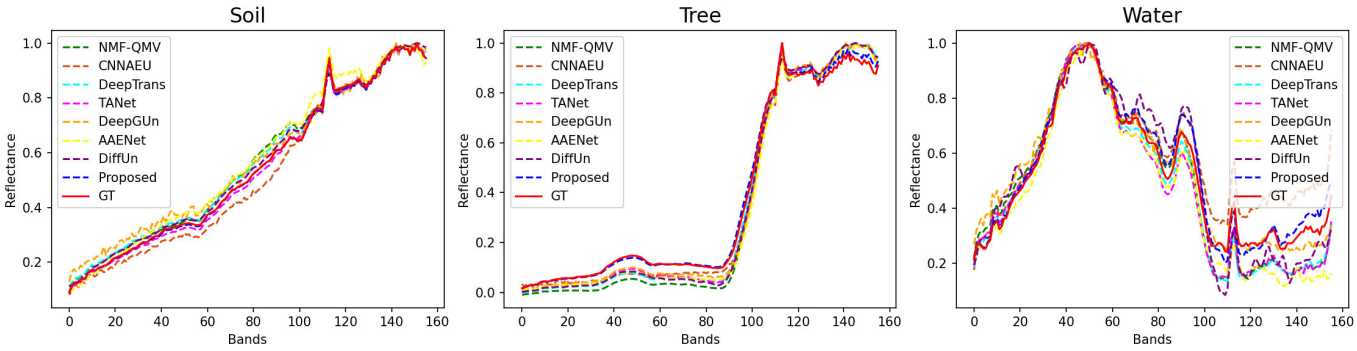


Fig. 8. Samson dataset: visual comparison of the endmembers obtained by the different unmixing techniques.

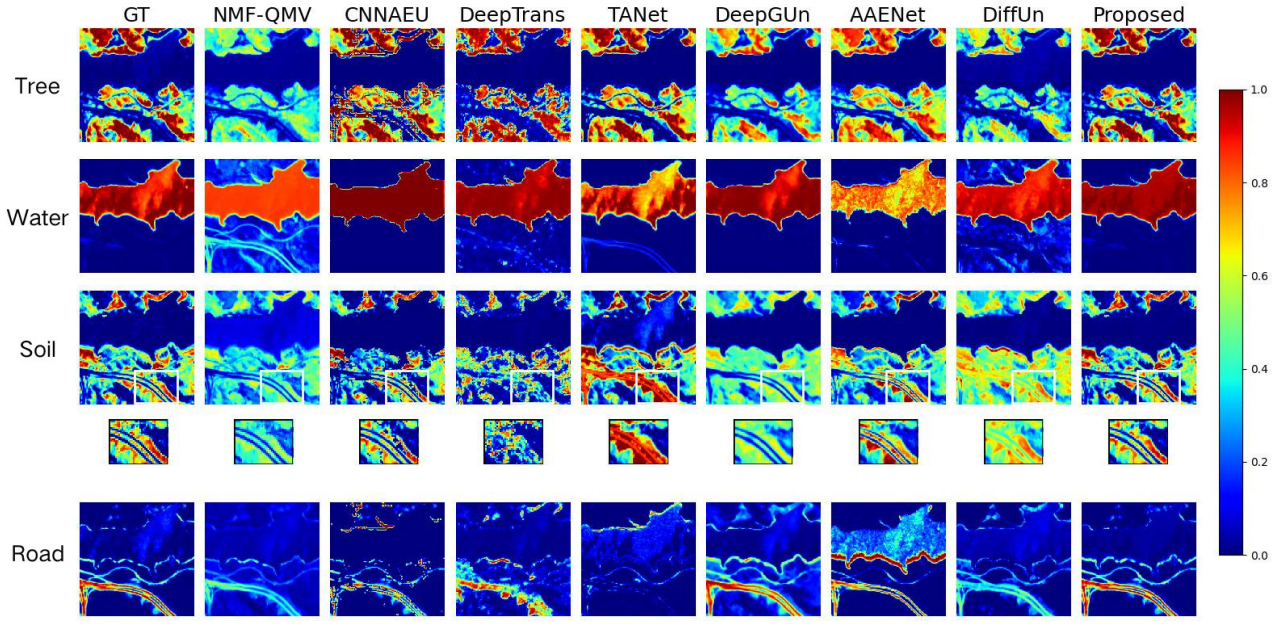


Fig. 9. Jasper Ridge dataset: visual comparison of the abundance maps obtained by the different unmixing techniques.

endmember spectral curves for all compared methods. Comparing the ground truth, we can find that our proposed method can provide a more accurate estimation. It can be observed that AAENet fails to adequately consider the sparsity

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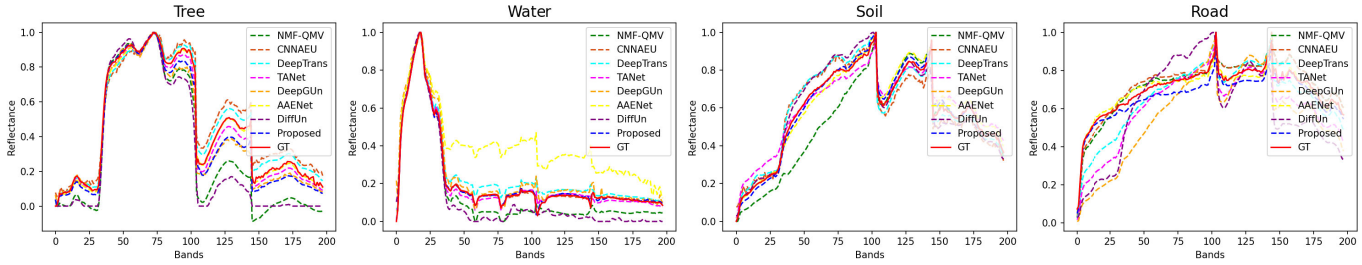


Fig. 10. Jasper Ridge dataset: visual comparison of the endmembers obtained by the different unmixing techniques.

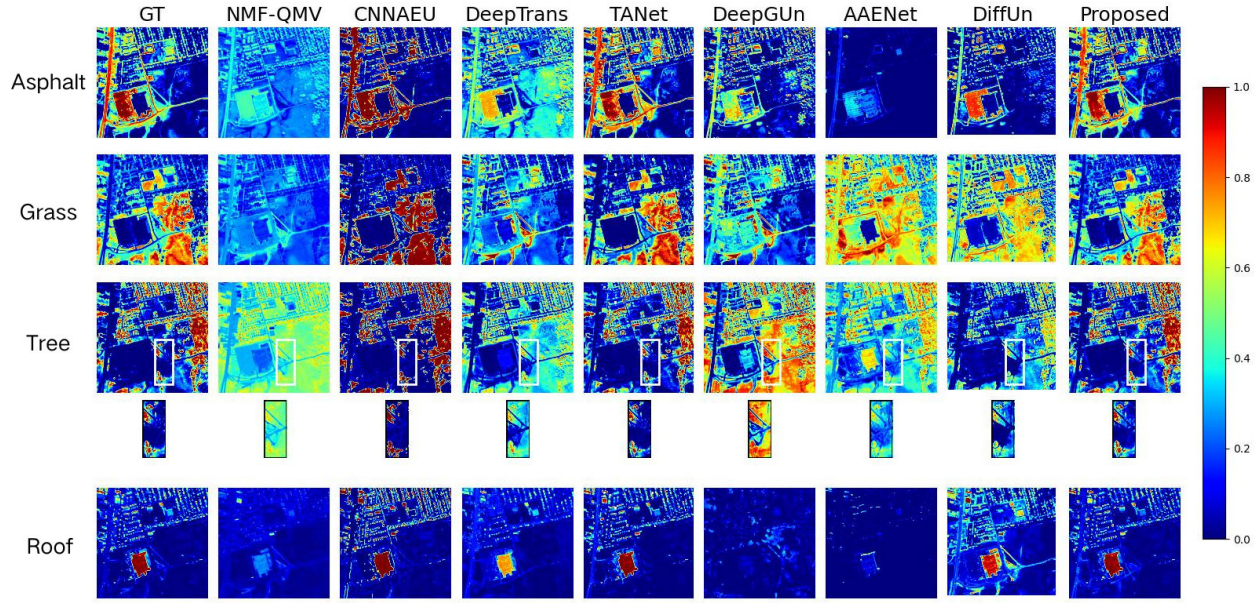


Fig. 11. Urban dataset: visual comparison of the abundance maps obtained by the different unmixing techniques.

of abundances, leading to unsatisfactory unmixing performance. The proposed method achieves the best or comparable results in most cases, which proves the effectiveness of the algorithm.

F. Experiments on Real Hyperspectral Data

1) Samson Dataset: Table III presents aRMSE, mSAD, and RE results for different unmixing methods on the Samson dataset, with the best results highlighted in bold. The estimated abundance maps and endmember spectral curves for all compared methods are shown in Figs. 7 and 8, respectively. The Samson dataset poses challenges for unmixing due to its spectral complexity and lack of pure pixels. NMF-QMV struggles with distinguishing soil and water, primarily because VCA relies on the unrealistic assumption of pure pixels. TANet suffers from over-smoothing, resulting in the loss of fine-grained details. In contrast, the proposed method shows sparse abundances for the region and captures fine details at region boundaries. For endmember extraction, our method achieves precise spectral signatures for soil and tree, and the extracted water endmember closely matches the reference with only minor deviations in highly mixed regions.

2) Jasper Ridge Dataset: Table IV presents aRMSE, mSAD, and RE results for different unmixing methods on the Jasper Ridge dataset, with the best results highlighted in bold. The estimated abundance maps and endmember spectral curves for all compared methods are shown in Figs. 9 and 10, respectively. Our method demonstrates superior performance in abundance estimation across both highly mixed regions (e.g., soil) and homogeneous areas (e.g., water). NMF-QMV and TANet overestimate the abundance of water in road-dominated pixels. The abundance maps generated by CNNAEU and DeepTrans lack smoothness and fail to capture certain fine details. As shown in the table, our method achieves the most accurate estimation of spectral signatures for the endmembers water, soil, and roads, outperforming all other compared methods. The extracted endmember spectra closely match the reference, with minimal SAD, demonstrating the effectiveness of our method.

3) Urban Dataset: Table V presents aRMSE, mSAD, and RE results for different unmixing methods on the Urban dataset, with the best results highlighted in bold. The estimated abundance maps and endmember spectral curves for all compared methods are shown in Figs. 11 and 12, respectively. The Urban dataset is highly complex due to its diverse materials and intricate spatial distributions. Our method achieves the lowest mSAD and aRMSE values among all the compared methods, demonstrating its effectiveness in both endmember extraction and abundance estimation. The second-best

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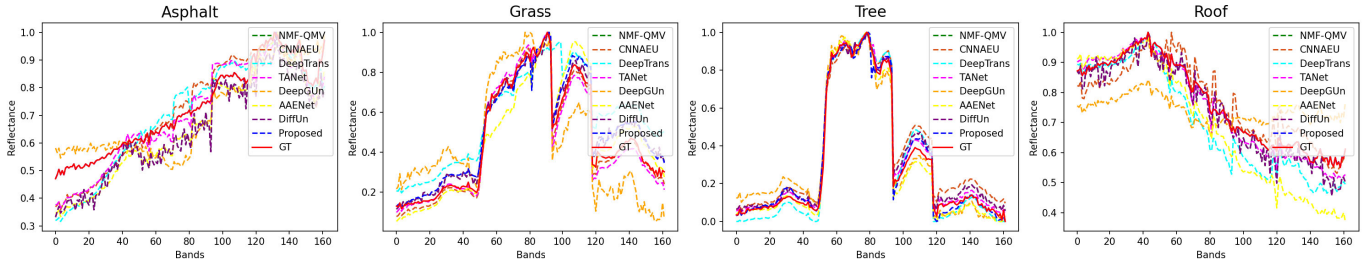


Fig. 12. Urban dataset: visual comparison of the endmembers obtained by the different unmixing techniques.

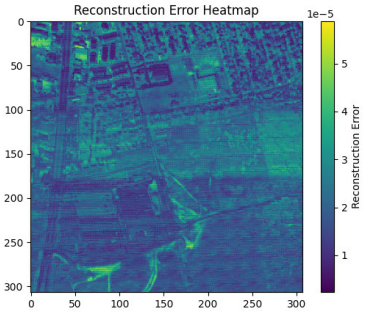


Fig. 13. Reconstruction error heatmap of the Urban dataset.

performance is observed with TANet, which also performs well but struggles in handling highly mixed pixels. The traditional method fails to effectively separate asphalt and grass due to their reliance on simplified assumptions, leading to significant errors. Additionally, methods like AAENet and CNNAEU are adversely affected by incorrect endmember extraction, resulting in poor abundance estimation. These results demonstrate the effectiveness of the proposed method in real complex scenes and confirm its enormous potential for practical applications.

Fig. 13 illustrates the reconstruction error heatmap of the Urban dataset. Higher errors are observed around complex structures like buildings and roads, indicating challenges in reconstructing mixed pixels. In contrast, uniform areas such as open fields exhibit lower errors, demonstrating better reconstruction performance in simpler regions.

4) Cuprite Dataset: Since the ground truth for the abundance maps in the Cuprite dataset is not available, three primary endmembers were selected for visualization. The abundance maps corresponding to these endmembers are presented in Fig. 14, while the spectral signatures of the extracted endmembers are shown in Fig. 15. The abundance maps display clear spatial distributions that reflect the geological structures of the region, and the spectral signatures align well with expected mineralogical features. These results highlight the practical applicability of the proposed method in mineral exploration and remote sensing.

G. Computation Cost

We evaluated the computational cost of the proposed method by analyzing the giga floating-point operations (GFLOPs) and running time on the Samson and Urban datasets. The results, presented in Table VI, illustrate the performance of multiple deep learning methods. Overall, diffusion-based methods, such as DiffUn and our proposed method, exhibit significantly higher computational costs compared to other approaches. However, despite the higher computational cost, diffusion-based methods often provide superior modeling capabilities, enabling them to better capture complex data patterns and improve overall performance. This tradeoff is particularly acceptable for high-precision tasks where more intricate data structures are required.

H. VCA Initialization Analysis

In the above experiments, VCA was used to initialize the endmembers. In this experiment, we investigate the impact of using VCA for endmember initialization on the performance of deep learning-based unmixing methods. To this end, we conducted comparative experiments where all methods were tested under two initialization strategies: VCA initialization and random initialization. The experimental results are presented in Table VII, with the best performance highlighted in bold.

The results show that on relatively simple datasets, some methods, such as DeepTrans, experience a significant performance drop when using random initialization compared to VCA initialization. This indicates that the lightweight and efficient VCA approach can provide a good initialization for endmembers, effectively guiding the optimization process. However, on more complex datasets, the performance gap between VCA initialization and random initialization narrows. This is because the simplistic nature of VCA limits its ability to handle complex datasets and extract high-quality endmembers. Additionally, our proposed method exhibits less dependence on VCA initialization compared to other methods, demonstrating its robustness and adaptability under different initialization conditions.

I. Ablation Study

To investigate the contribution of each module in our method, we conduct a series of ablation experiments. The experimental results are presented in Table VIII, with the best results highlighted in bold. In the absence of the denoising guidance generator, we replace it with a fully connected layer to take its place in the original network and perform dimensionality reduction. Removing the denoising guidance generator significantly reduces the model's ability to extract high-level spectral–spatial features, leading to poorer

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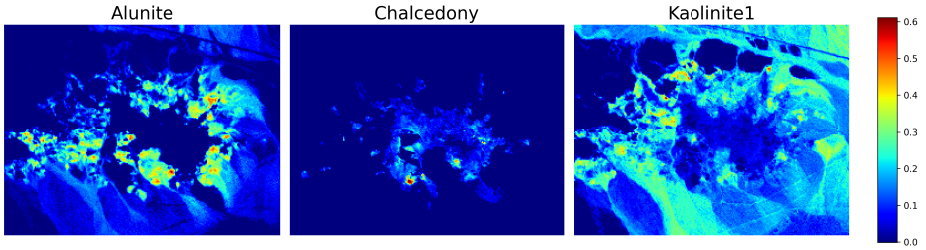


Fig. 14. Cuprite dataset: visualization of the abundance maps obtained by the proposed method.

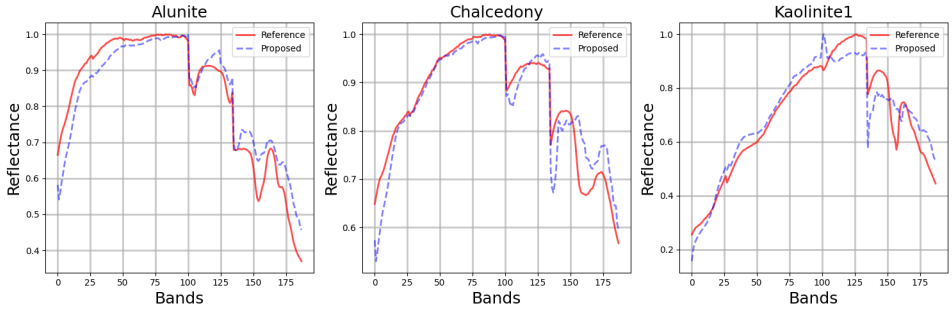


Fig. 15. Cuprite dataset: visualization of the endmembers obtained by the proposed method.

TABLE III
QUANTITATIVE COMPARISONS ON THE SAMSON DATASET

	NMF-QMV	CNNAEU	DeepTrans	TANet	DeepGUn	AAENet	DiffUn	Proposed
Soil	0.1973	0.1870	0.0905	0.2229	0.0612	0.1831	0.0945	**0.0461**
Tree	0.1451	0.2030	0.0537	0.1059	0.0648	0.1298	0.1094	**0.0395**
Water	0.2037	0.1775	0.0569	0.2844	0.0504	0.2407	0.0985	**0.0278**
aRMSE	0.1839	0.1895	0.0691	0.2154	0.0591	0.1900	0.1008	**0.0386**
mSAD	0.0946	0.0867	0.0788	0.0738	0.0688	0.0934	0.0832	**0.0222**
RE(e^{-2})	0.0054	0.0916	0.0497	0.3649	0.4467	0.5469	0.0102	**0.0049**

TABLE IV
QUANTITATIVE COMPARISONS ON THE JASPER RIDGE DATASET

	NMF-QMV	CNNAEU	DeepTrans	TANet	DeepGUn	AAENet	DiffUn	Proposed
Tree	0.2047	0.2837	0.1593	0.0834	0.1331	0.0900	0.1691	**0.0334**
Water	0.1838	0.1587	0.1080	0.0741	**0.0441**	0.2072	0.0862	0.0504
Soil	0.1468	0.1744	0.2188	0.2133	0.1451	0.1193	0.1772	**0.0493**
Road	0.1246	0.2064	0.1784	0.2294	0.1152	0.2481	0.0969	**0.0506**
aRMSE	0.1679	0.2114	0.1709	0.1662	0.1162	0.1781	0.1378	**0.0467**
mSAD	0.1832	0.1182	0.0891	0.0837	0.1541	0.1273	0.1995	**0.0325**
RE(e^{-2})	0.0947	3.3151	0.2007	4.8613	5.5975	0.2931	**0.0621**	0.0941

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TABLE V
QUANTITATIVE COMPARISONS ON THE URBAN DATASET

	NMF-QMV	CNNAEU	DeepTrans	TANet	DeepGUn	AAENet	DiffUn	Proposed
Asphalt	0.3098	0.2060	0.3500	0.1006	0.2648	0.3734	0.2019	**0.0765**
Grass	0.3164	0.2229	0.3280	0.1191	0.2868	0.3210	0.1532	**0.1135**
Tree	0.3197	0.1597	0.2413	**0.0706**	0.2694	0.2587	0.1076	0.0817
Roof	0.1940	0.1710	0.1257	0.0536	0.2442	0.2158	0.1121	**0.0476**
aRMSE	0.2898	0.1914	0.2757	0.0892	0.2460	0.2983	0.1437	**0.0832**
mSAD	0.1551	0.0806	0.1206	0.0543	0.1677	0.1072	0.0725	**0.0427**
RE(e^{-2})	0.0728	0.9609	0.0448	1.0941	1.3073	0.5632	**0.0428**	0.0668

TABLE VI
COMPUTATIONAL COST OF ALL DEEP LEARNING METHODS ON DIFFERENT DATASETS

Dataset	Evaluation Metrics	CNNAEU	DeepTrans	TANet	DeepGUn	AAENet	DiffUn	Proposed
Samson	GFlops	**0.23G**	0.82G	0.86G	3.11G	8.82G	8.43G	8.61G
	Running Time	215s	**180s**	253s	196s	287s	439s	358s
Urban	GFlops	**0.28G**	25.4G	12.08G	33.60G	36.25G	42.11G	44.07G
	Running Time	361s	409s	611s	**401s**	564s	758s	843s

TABLE VII
COMPARISON OF UNMIXING PERFORMANCE WITH AND WITHOUT VCA INITIALIZATION OF ALL DEEP LEARNING METHODS ON DIFFERENT DATASETS

Dataset	VCA	Evaluation Metrics	CNNAEU	DeepTrans	TANet	DeepGUn	AAENet	DiffUn	Proposed
Samson	✓	aRMSE	0.1895	0.0691	0.2154	0.0591	0.1900	0.1008	**0.0386**
		mSAD	0.0867	0.0788	0.0738	0.0688	0.0934	0.0832	**0.0222**
	×	aRMSE	0.1984	0.1619	0.2233	0.0654	0.2057	0.1268	**0.0494**
		mSAD	0.1317	0.1478	0.1512	0.0935	0.1338	0.1017	**0.0669**
Urban	✓	aRMSE	0.1914	0.2757	0.0892	0.2460	0.2983	0.1437	**0.0832**
		mSAD	0.0806	0.1206	0.0543	0.1677	0.1072	0.0725	**0.0427**
	×	aRMSE	0.1962	0.3042	0.1528	0.2545	0.3011	0.1552	**0.0912**
		mSAD	0.1246	0.1515	0.0949	0.1714	0.1157	0.0838	**0.0516**

TABLE VIII
QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS

Module			Synthetic dataset		Samson		Jasper Ridge		Urban	
Denoising Guidance Generator	Noise-to-Abundance Transformer	Stage-wise Training	aRMSE	mSAD	aRMSE	mSAD	aRMSE	mSAD	aRMSE	mSAD
×	✓	✓	0.0474	0.0256	0.0976	0.0546	0.0568	0.0827	0.1075	0.1101
✓	×	✓	0.0478	0.0584	0.0859	0.0855	0.1292	0.1149	0.1181	0.1749
✓	✓	×	0.0527	0.0319	0.0458	0.0699	0.1186	0.1061	0.1149	0.1381
✓	✓	✓	**0.0256**	**0.0092**	**0.0386**	**0.0222**	**0.0467**	**0.0325**	**0.0832**	**0.0427**

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abundance estimation. Without the noise-to-abundance Transformer, the model fails to achieve the same level of precision in abundance estimation and endmember extraction, resulting in higher reconstruction errors. The stage-wise training proves particularly effective in guiding model training on complex datasets, ensuring stable training and preventing overfitting. These results demonstrate that each module plays a crucial role in the framework and their combined use is essential for achieving robust and accurate hyperspectral unmixing.

V. Conclusion

In this article, we propose a novel unsupervised hyperspectral unmixing method based on diffusion models, comprising three key modules: a denoising guidance generator to extract meaningful features and provide robust guidance, a noise-to-abundance Transformer to convert noise into abundance maps, and an abundance evaluation module to ensure the quality and reliability of the generated abundance maps. Additionally, we adopt a stage-wise training strategy to stabilize the optimization process and improve training efficiency. The experiments on synthetic and real hyperspectral data demonstrate that the proposed algorithm can achieve better unmixing performance compared with the state-of-the-art methods.

Future work could address the limitations of using a linear decoder in the abundance evaluation module, which sometimes results in suboptimal reconstruction accuracy due to the nonlinear nature of real-world spectral mixing. This could be achieved by exploring the incorporation of nonlinear decoders within the module to better capture these complex interactions. Additionally, our method can be adapted to handle spectral variability by applying superpixel segmentation before patch-based processing of the noise-to-abundance Transformer, further enhancing its applicability to real-world scenarios.

References

[1] M. Berman, P. M. Connor, L. B. Whitbourn, D. A. Coward, B. G. Osborne, and M. D. Southan, "Classification of sound and stained wheat grains using visible and near infrared hyperspectral image analysis," *J. Near Infr. Spectrosc.*, vol. 15, no. 6, pp. 351–358, Dec. 2007.

[2] A. Gowen, C. Odonnell, P. Cullen, G. Downey, and J. Frias, "Hyperspectral imaging—An emerging process analytical tool for food quality and safety control," *Trends Food Sci. Technol.*, vol. 18, no. 12, pp. 590–598, Dec. 2007.

[3] W. Wei, L. Zhang, Y. Jiao, C. Tian, C. Wang, and Y. Zhang, "Intracluster structured low-rank matrix analysis method for hyperspectral denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 866–880, Feb. 2019.

[4] L. Zhang, W. Wei, C. Bai, Y. Gao, and Y. Zhang, "Exploiting clustering manifold structure for hyperspectral imagery super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 12, pp. 5969–5982, Dec. 2018.

[5] L. Wang, X. Zhang, J. Zhang, H. Dong, H. Meng, and L. Jiao, "Pixel-to-abundance translation: Conditional generative adversarial networks based on patch transformer for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 5734–5749, 2024.

[6] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[7] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," *Proc. SPIE*, vol. 3753, pp. 266–277, Oct. 1999.

[8] J. M. P. Nascimento and J. M. Bioucas-Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[9] D. C. Heinz, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.

[10] D. Zhang, Z.-H. Zhou, and S. Chen, "Non-negative matrix factorization on kernels," in *Proc. PRICAI Trends Artif. Intell., 9th Pacific Rim Int. Conf. Artif. Intell. Guilin*, Jan. 2006, pp. 404–412.

[11] A. Halimi, Y. Altmann, N. Dobigeon, and J.-Y. Tourneret, "Nonlinear unmixing of hyperspectral images using a generalized bilinear model," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4153–4162, Nov. 2011.

[12] B. Rasti, A. Zouaoui, J. Mairal, and J. Chanussot, "Image processing and machine learning for hyperspectral unmixing: An overview and the HySUPP Python package," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517631.

[13] J. S. Bhatt and M. V. Joshi, "Deep learning in hyperspectral unmixing: A review," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 2189–2192.

[14] V. S. Deshpande, and J. S. Bhatt, "A practical approach for hyperspectral unmixing using deep learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[15] A. Zouaoui, G. Muhawenayo, B. Rasti, J. Chanussot, and J. Mairal, "Entropic descent archetypal analysis for blind hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 32, pp. 4649–4663, 2023.

[16] L. Wan, T. Chen, A. Plaza, and H. Cai, "Hyperspectral unmixing based on spectral and sparse deep convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11669–11682, 2021.

[17] B. Palsson, M. O. Ulfarsson, and J. R. Sveinsson, "Convolutional autoencoder for spectral–spatial hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 535–549, Jan. 2020.

[18] P. Ghosh, S. K. Roy, B. Koirala, B. Rasti, and P. Scheunders, "Hyperspectral unmixing using transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535116.

[19] L. P. Cinelli, M. A. Marins, E. A. B. Da Silva, and S. L. Netto, "Variational autoencoder," in *Variational Methods for Machine Learning With Applications to Deep Networks*. Cham, Switzerland: Springer, 2021, pp. 111–149.

[20] I. J. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[21] R. A. Borsoi, T. Imbiriba, and J. C. M. Bermudez, "Deep generative endmember modeling: An application to unsupervised spectral unmixing," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 374–384, 2020.

[22] Q. Jin, Y. Ma, F. Fan, J. Huang, X. Mei, and J. Ma, "Adversarial autoencoder network for hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 8, pp. 4555–4569, Aug. 2021.

[23] I. Higgins et al., "Beta-VAE: Learning basic visual concepts with a constrained variational framework," in *Proc. ICLR (Poster)*, Apr. 2017, pp. 1–14.

[24] F.-A. Croitoru, V. Hondru, R. T. Ionescu, and M. Shah, "Diffusion models in vision: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10850–10869, Sep. 2023.

[25] Y. Zhang, B. Hartl, H. Hazan, and M. Levin, "Diffusion models are evolutionary algorithms," 2024, *arXiv:2410.02543*.

[26] Y. Yu, E. Pan, Y. Ma, X. Mei, Q. Chen, and J. Ma, "UnmixDiff: Unmixing-based diffusion model for hyperspectral image synthesis," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524018.

[27] K. Deng, Y. Qian, J. Nie, and J. Zhou, "Diffusion-model-based hyperspectral unmixing using spectral prior distribution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519716.

[28] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," 2020, *arXiv:2010.02502*.

[29] J. Chen, M. Zhao, X. Wang, C. Richard, and S. Rahardja, "Integration of physics-based and data-driven models for hyperspectral image unmixing: A summary of current methods," *IEEE Signal Process. Mag.*, vol. 40, no. 2, pp. 61–74, Mar. 2023.

[30] Q. Cheng, B. Du, L. Zhang, and R. Liu, "ANSGA-III: A multiobjective endmember extraction algorithm for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 2, pp. 700–721, Feb. 2019.

[31] T. Amit, T. Shaharbany, E. Nachmani, and L. Wolf, "SegDiff: Image segmentation with diffusion probabilistic models," 2021, *arXiv:2112.00390*.

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[32] D. Baranchuk, I. Rubachev, A. Voynov, V. Khrulkov, and A. Babenko, "Label-efficient semantic segmentation with diffusion models," 2021, *arXiv:2112.03126*.

[33] J. Austin, D. D. Johnson, J. Ho, D. Tarlow, and R. V. D. Berg, "Structured denoising diffusion models in discrete state-spaces," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2021, pp. 17981–17993.

[34] E. Hoogeboom, D. Nielsen, P. Jaini, P. Forré, and M. Welling, "Argmax flows and multinomial diffusion: Learning categorical distributions," in *Proc. NIPS*, 2021, pp. 12454–12465.

[35] O. Avrahami, D. Lischinski, and O. Fried, "Blended diffusion for text-driven editing of natural images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 18208–18218.

[36] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen, "Hierarchical text-conditional image generation with CLIP latents," 2022, *arXiv:2204.06125*.

[37] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NIPS*, vol. 33. Vancouver, BC, Canada: Curran Associates, 2020, pp. 6840–6851.

[38] L. Zhang and M. Agrawala, "Adding conditional control to text-to-image diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 3836–3847.

[39] H. Liu et al., "AudioLDM: Text-to-audio generation with latent diffusion models," 2023, *arXiv:2301.12503*.

[40] J. Ho, T. Salimans, A. Gritsenko, W. Chan, M. Norouzi, and D. J. Fleet, "Video diffusion models," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 8633–8646. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2022/file/39235c56aef13fb05a6adc95eb9d8d66-Paper-Conference.pdf

[41] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in *Proc. NIPS*, vol. 34, 2021, pp. 8780–8794.

[42] Z. Zou, L. Chen, X. Jiang, and A. M. Zoubir, "Exploiting generative diffusion prior with latent low-rank regularization for image inpainting," *IEEE Signal Process. Lett.*, vol. 31, pp. 2335–2339, 2024.

[43] W. Peebles and S. Xie, "Scalable diffusion models with transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 4195–4205.

[44] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[45] A. M. Baldridge, S. J. Hook, C. I. Grove, and G. Rivera, "The ASTER spectral library version 2.0," *Remote Sens. Environ.*, vol. 113, no. 4, pp. 711–715, Apr. 2009.

[46] Y. Zhou, A. Rangarajan, and P. D. Gader, "A spatial compositional model for linear unmixing and endmember uncertainty estimation," *IEEE Trans. Image Process.*, vol. 25, no. 12, pp. 5987–6002, Dec. 2016.

[47] B. Palsson, J. Sigurdsson, J. R. Sveinsson, and M. O. Ulfarsson, "Hyperspectral unmixing using a neural network autoencoder," *IEEE Access*, vol. 6, pp. 25646–25656, 2018.

[48] S. Ozkan, B. Kaya, and G. B. Akar, "EndNet: Sparse AutoEncoder network for endmember extraction and hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 482–496, Jan. 2019.

[49] Q. Qu, N. M. Nasrabadi, and T. D. Tran, "Subspace vertex pursuit: A fast and robust near-separable nonnegative matrix factorization method for hyperspectral unmixing," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 6, pp. 1142–1155, Sep. 2015.

[50] W. Wang, Y. Qian, and Y. Y. Tang, "Hypergraph-regularized sparse NMF for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 681–694, Feb. 2016.

[51] L. Zhuang, C.-H. Lin, M. A. Figueiredo, and J. M. Bioucas-Dias, "Regularization parameter selection in minimum volume hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9858–9877, Dec. 2019.

[52] Q. Jin, Y. Ma, X. Mei, and J. Ma, "TANet: An unsupervised two-stream autoencoder network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506215.

[53] Q. Du, N. Raksuntorn, N. H. Younan, and R. L. King, "End-member extraction for hyperspectral image analysis," *Appl. Opt.*, vol. 47, no. 28, pp. F77–F84, 2008.

[54] S. Vijayashekhar, J. S. Bhatt, and B. Chattopadhyay, "Virtual dimensionality of hyperspectral data: Use of multiple hypothesis testing for controlling type-I error," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2974–2985, 2020.

[55] C.-I. Chang, "A review of virtual dimensionality for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1285–1305, Apr. 2018.

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